Photonics - Spring 2023
Review Questions

(1) A ray of red light in air is incident on the surface of an unknown non-magnetic dielectric material at an angle of 39°. It was found that the refracted ray makes an angle of 27°. Find the refractive index of this material

(2) Compare the focal length of a lens surrounded by air and the same lens surrounded by a liquid of refractive index 1.6. Explain your answer

(3) Consider a 5-mW He-Ne laser at wavelength 632.8 nm. How many photons are generated every 1 ps?

(4) Explain why an optical cavity resonator is needed in a semiconductor laser. Is it possible to have a single-mode laser? How?

(5) Discuss the main differences between edge and surface light emitting diodes. What are the best applications for each type?

(6) Consider a semiconductor Fabry-Perot optical cavity of length 200 microns with end mirrors of 80% reflectance. The refractive index of the semiconductor is 3.7. Calculate the cavity mode nearest to the free space wavelength of 1300 nm. Also, calculate the spectral width (in Hz).

(7) Calculate the thickness of a Ge crystal layer needed for absorbing 90% of incident radiation at 1550 nm. Repeat for $In_{0.53}Ga_{0.47}As$.

(8) A GaAs LED emits at 860 nm. It is connected to a 9 V battery and draws a current of 20 mA. If the internal efficiency of the LED is 0.85, calculate the generated photon flux